

Use of Exoskeleton Chair for Supporting Posture Change in VR Experience

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Figure 1: (Left) First-person view of the custom VR scooter task used in the experiment. (Right) A participant using the posture-supportive exoskeleton chair while wearing a head-mounted display and hand controllers. Participants wore straps attached to the feet, legs, and waist, allowing them to naturally transition between sitting and standing positions during gameplay, with the exoskeleton supporting their body weight.

ABSTRACT

This study investigates the impact of a posture-supportive device on user experience in virtual reality (VR), focusing on fatigue, cybersickness, overall experience, and task performance. Using an adjustable exoskeleton chair, participants engaged in a custom-built VR game that required posture transitions between sitting and standing while completing interactive tasks. While the Normal Chair (N) condition resulted in the highest task performance, participants in the Without Chair (W) condition reported a more positive game experience. Physiological signals, including heart rate and electrodermal activity, indicated that the posture-supportive exoskeleton (E) condition resulted in reduced fatigue compared to the without-chair (W) condition.

Index Terms: Virtual Reality, Exoskeleton Chair, Posture Transition, Immersion, Cybersickness, User Experience

Index Terms: Human-centered computing—Virtual reality; Computing methodologies—Perception; Human-centered computing—User studies; Human-centered computing—User interface design;

1 INTRODUCTION

Posture plays a critical role in VR comfort, task performance, and immersion [8]. Instability and visual-vestibular conflict are known contributors to motion sickness [7, 2], while realism and motor engagement can enhance presence and user experience [6]. To advance VR experiences, we conduct a preliminary investigation into the use of a lower-body exoskeleton chair for supporting natural embodied interaction in tasks that require frequent posture transitions. We compare fatigue, cybersickness, and task performance across three conditions, use of Normal Chair (N), Exoskeleton (E), and Without Chair (W), during a VR scooter-riding task.

2 METHODS

2.1 Experimental Platform

A Unity-based VR scooter-riding game was developed in which users collected floating coins using their handcontroller (See Fig-

ure 1 (Left)) while controlling speed through posture (sitting or standing) (See Figure 1 (Right)). Standing increased scooter velocity, while sitting reduced it. The environment was presented through an HTC VIVE Pro Eye HMD with handheld controllers, and all sessions were executed on a desktop PC running Windows 10 (AMD Ryzen 7 3700X 8-core processor at 3.6 GHz, 16 GB RAM, NVIDIA GeForce RTX 3060). We used the EmotiBit¹ for the physiology signal data collection during the experiment. We used a passive exoskeleton chair provided semi-seated support. Adjustable straps and locking joints allowed participants to alternate between upright and partially seated positions without powered actuation (See Figure 1 (Right)). The device supports up to 150 kg and weighs approximately 1.8 kg. In the Normal Chair (N) condition, participants remained fully seated in a standard chair, while in the Without Chair (W) condition, they stood for the entire trial with no support.

2.2 Study Design, Dependent Variables, and Protocol

The study employed a within-subject Latin-square design with twelve participants (ages 20–26), with a mean height of 172.3 cm (SD = 8.0) and mean weight of 71.3 kg (SD = 9.6), ensuring all posture conditions were experienced once by each participant. Each trial lasted 300 seconds, and total experiment time per participant was approximately 25 minutes, including calibration and breaks. We collected subjective measures of cybersickness using the nausea subscale of the Simulator Sickness Questionnaire (SSQ) [3] and fatigue using a modified version of the Game Experience Questionnaire (GEQ) [4]. Performance measures included total coins collected and mean scooter speed, reflecting task efficiency and posture-dependent movement control. We also collected physiological signals, including heart rate (HR), electrodermal activity (EDA), blood volume pulse (BVP), and skin temperature. Subjective responses related to the gaming experience were gathered using GEQ [4]. After providing informed consent, participants completed a baseline questionnaire and were fitted with the HMD and controllers. For the Exoskeleton condition, experimenters assisted with device fitting and posture calibration. Participants completed the gameplay per posture condition, with break intervals between trials. After each trial, participants reported fatigue and cybersickness before proceeding to the next condition.

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¹<https://docs.openbci.com/ThirdParty/EmotiBit/EmotiBitGuide/>

Table 1: Statistical results for game experience, performance, physiology, and cybersickness across conditions (E: Exoskeleton, N: Normal Chair, W: Without Chair; B: Baseline). **p < 0.05, *p < 0.1. **Bold** indicates the condition with the most notable mean.

Category	Dvs	One-way ANOVA	Effect Size	E	N	W	B
Game Experience	Negative	$F(2, 22) = 1.18, p = 0.326$	0.1	2.63 (0.69)	2.38 (0.58)	2.57 (0.58)	
	Positive	$F(2, 22) = 0.21, p = 0.809$	0.02	5.93 (0.51)	5.85 (0.58)	5.95 (0.7)	
	Competence	$F(2, 22) = 2.53, p = 0.10*$	0.19	5.48 (0.81)	5.16 (1.0)	5.71 (0.86)	
	Flow	$F(2, 22) = 2.84, p = 0.08*$	0.21	5.43 (1.03)	5.35 (0.99)	5.93 (1.09)	
	Engagement	$F(2, 22) = 1.15, p = 0.334$	0.1	6.04 (0.59)	5.95 (0.84)	6.27 (0.74)	
	Fatigue	$F(2, 22) = 27.20, p = 0.001**$	0.71	4.38 (1.36)	2.42 (1.08)	5.25 (0.76)	
Game Performance	Collected Coins	$F(2, 22) = 6.47, p = 0.001**$	0.37	266.5 (70.4)	327.75 (71.86)	322.33 (89.23)	
	Average Speed	$F(2, 22) = 3.09, p = 0.066*$	0.22	12.07 (2.77)	14.11 (2.46)	12.91 (2.05)	
Physiology Data	EDA	$F(2, 22) = 396.19, p = 0.001**$	0.12	3.45 (3.28)	1.81 (1.83)	4.47 (3.68)	
	HR	$F(2, 22) = 61.90, p = 0.001**$	0.06	86.12 (7.92)	87.43 (11.31)	92.58 (15.94)	
	BVP	$F(2, 22) = 0.078, p = 0.925$	0.06	48.82 (0.89)	48.54 (0.31)	48.34 (1.04)	
	Temp	$F(2, 22) = 24.17, p = 0.001**$	0.01	33.94 (2.03)	34.19 (1.26)	34.23 (1.24)	
Cybersickness	Nausea	$F(3, 44) = 12.65, p = 0.001**$	0.54	100.96 (7.05)	77.91 (9.82)	93.81 (13.39)	76.32 (7.05)
	Oculomotor	$F(3, 44) = 3.11, p = 0.064*$	0.22	60.01 (12.29)	52.43 (7.55)	61.27 (9.94)	51.8 (7.81)
	Disorientation	$F(3, 44) = 1.0, p = 0.384$	0.09	98.60 (25.50)	88.16 (12.36)	91.64 (12.53)	84.68 (4.02)
	Total Score	$F(3, 44) = 6.56, p = 0.001**$	0.31	970.80 (181.11)	817.18 (86.42)	922.74 (97.39)	795.86 (40.23)

3 RESULTS

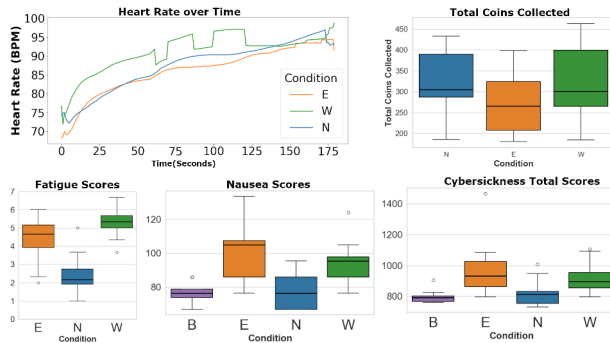


Figure 2: Key results across posture conditions in VR, showing heart rate, task performance, fatigue, nausea, and total cybersickness for B (Baseline), E (Exoskeleton), N (Normal chair), and W (Without chair).

We provide the inferential statistical results in Table 1 and box plots of the key findings in Figure 2.

Cybersickness and Fatigue The nausea scores lowest in N, intermediate in W, and highest in E. The results showed the lowest fatigue in the Normal Chair (N) condition, moderate fatigue in the Exoskeleton (E) condition, and the highest fatigue in the Without Chair (W) condition.

Performance Task performance was highest in the Normal Chair (N) condition, followed closely by Without Chair (W), with the lowest in Exoskeleton (E).

Physiological Signal Physiological measures aligned with subjective fatigue reports. Heart rate (HR) and electrodermal activity (EDA) were highest in W, reflecting greater physical exertion during unsupported standing, lowest in N, and intermediate in E.

Experience The Without Chair (W) condition indicated the highest Competence and Flow in terms of game experience, followed by the Exoskeleton (E) condition, with the lowest scores observed in the Normal Chair (N) condition.

4 DISCUSSION

The results suggest a trade-off in posture support for VR. The exoskeleton appeared to reduce physical fatigue compared to unsupported standing, consistent with previously reported posture-transition support benefits of wearable devices [1]. However, it was

also associated with a slight increase in nausea, which may be related to subtle instability or sensory conflict from the passive exoskeleton chair. Task performance and overall comfort were highest with the Normal Chair, while unsupported standing imposed the greatest physiological load [5]. These findings indicate that wearable posture solutions can provide targeted benefits, though they may not improve all aspects of the VR experience.

5 CONCLUSION

This study offers a preliminary look at lower-body exoskeleton chairs in VR, focusing on their potential to support posture and task engagement. The observations suggest that wearable posture-support mechanisms should be considered in relation to the sensory and cognitive demands of immersive environments. Future work could explore adaptive support strategies, improved stabilization techniques, and longer exposure periods to assess their suitability in dynamic VR tasks.

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